

Greater China Industrials

Robotics company visit takeaways: world action model, data and dexterous hand

Industry Overview

More diversified application of embodied AI in 2026

We visited six embodied AI companies in July, including humanoid & service robot OEMs (Galaxea AI, OneRobotics, Yunji,) and component makers (Linkerbot, Orbbec, PaXini). We now see more diversified applications of embodied AI, expanding from mostly education, R&D and data collection in 2025 to industrial, logistics, retail, services and household scenario in 2026. Most robotics companies now see industrial and logistics becoming the major application scenario of embodied AI in the coming few years, while some companies are also actively expanding in household and services market, in order to capture a potentially larger market in the long run.

World action model on the rise

In 2026, many embodied AI companies are shifting towards world action model (WAM), which combines the VLA model with a world model (such as OneRobotics' OneModel 1.7 and Galaxea AI's Fast-WAM). Compared to a pure VLA model, WAM can better learn the physical laws of the real world, with higher generalization capability and performs well in long-period tasks. With the aid of WAM, robotics companies are now able to perform more complex tasks lasting several minutes, such as washing and folding clothes, preparing meals and washing dishes.

Data to become the new battlefield in 2026

On the data front, most companies now use a combination of simulated data and real-world data, with an increasing emphasis on real-world data given its importance in training high-quality embodied AI models. While the cost of obtaining real-world data remains high compared to simulated data, robotics companies are able to lower the data collection costs with UMI and Egocentric data (c.90% cheaper than teleoperation). Moreover, the higher costs in obtaining real-world data could be offset by the efficiency gain in model training process, given the higher data quality. Leading robotics & component companies now target to collect >1mn hours of real-world data spanning across industrial, logistics, service and household scenarios. Apart from data volume, data quality and diversity are also key consideration of embodied AI developers.

Dexterity hand: falling cost and improved dexterity

The ASP of dexterous hands has declined rapidly in the past two years, thanks to scale effect, improved production process, and modularized design. Dexterous hand makers are also launching high-end models with high dexterity (>20 DoF), multimodal sensory, long durability and lightweight design. Tactile sensors have become a must-have for dexterous hands, which provides high-quality data for VTLA models. Leading dexterity hand companies have also engaged in the development of VTLA models and data collection solutions (data factory/data gloves), given the crucial role of dexterous hand in data collection and manipulation activities.

11 August 2026

Equity
Greater China
Industrials

Ming Hsun Lee, CFA >>
Research Analyst
Merrill Lynch (Hong Kong)
minghsun.lee@bofa.com

Yikai Liu, CFA >>
Research Analyst
Merrill Lynch (Hong Kong)

Summer Wang, CFA >>
Research Analyst
Merrill Lynch (Hong Kong)

Fiona Liang >>
Research Analyst
Merrill Lynch (Hong Kong)

HR: humanoid robot

AI: artificial intelligence

VLA: vision-language-action

VTLA: vision-tactile-language-action

DoF: degree of freedom

UMI: universal manipulation interface

OEM: original equipment
manufacturer

[Greater China Industrials \(H\): Who Makes the Humanoid Robot \(in 2026\)? Locomotion first, manipulation follows](#)

>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.

Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.

BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Refer to important disclosures on page 11 to 12.

Key takeaways – Robot OEMs

Galaxea AI (private)

Galaxea is an embodied AI company focused on developing foundation models and intelligent robot hardware. Its core products include wheeled robots (R1/R1 Pro), dual-arm mobile robots (R1 Lite), and robotic arms (A1), supported by self-developed AI algorithms, data platforms, and world models. It mainly sells its robots to developers and industrial & commercial customers.

Galaxea has launched and open-sourced its new-generation VLA base model, G0.5, with strong zero-shot generalization capability. It has also launched its world action model, Fast-WAM, based on a world model coupled with a dual-system VLA model for full-body control. On the data front, Galaxea primarily uses real-world data (a combination of teleoperation, UMI and Egocentric data) for model training. According to Galaxea, the cost of UMI and Egocentric data is around one-tenth that of teleoperation data, which could help lower the overall cost of data collection. Galaxea has also announced that it has formed a JV with Beijing Economic-Technological Development Area, focusing on data collection in logistics, restaurants, hotels and retail scenarios. It targets to collect >1mn hours of real-world data in 2026.

Exhibit 1: Galaxea R1 wheeled robot

Galaxea AI's products include wheeled robots, mobile robots & robotic arms



Source: Galaxea AI

BofA GLOBAL RESEARCH

Exhibit 2: Galaxea R1 Lite dual-arm mobile robot

R1 Lite focuses on data collection



Source: Galaxea AI

BofA GLOBAL RESEARCH

OneRobotics (6600 HK)

OneRobotics is a leading provider of home service robots in China. It has built a diversified product portfolio around its “One Brain, Multiple Embodiments” embodied AI platform, covering three major product lines, including: (1) Home service robots (Onero), designed to automate everyday household tasks such as cleaning, home management, and elderly care; (2) Tennis robots (Acemate), focusing on interactive training and fitness applications, including AI-powered sports coaching; and (3) Companion robots (SwitchBot), which enhances emotional engagement through natural interaction, entertainment, and personalized companionship.

The company expects its revenue to reach RMB1.4-1.5bn in 2026 (+>50% YoY) and deliver a 2026-28 revenue CAGR of no less than 50%, supported by its established SwitchBot portfolio and new embodied AI products (including Acemate, Onero and its new companion robots). It recently launched OneModel 1.7, featuring a single world action model that integrates world model with the VLA architecture. On the data front, the company is building a data collection center in the Greater Bay Area and targets to collect 1mn hours of robotics data in 2026 through teleoperation, UMI and Egocentric data collection. Its data collection mainly focuses on household scenarios, including washing and folding clothes, preparing meals and washing dishes. Its embodied AI model can now achieve c.70% accuracy rate in these scenarios.

Exhibit 3: OneRobotics' Acemate

Acemate enables AI-powered sports coaching



Source: OneRobotics

BofA GLOBAL RESEARCH

Exhibit 4: OneRobotics' Onero

Onero is a home service robot for every household tasks



Source: OneRobotics

BofA GLOBAL RESEARCH

Yunji (2670 HK)

Yunji is a leading company for robotic intelligent agents in China. Yunji leverages intelligent agents to deliver adaptive, scalable, consumer-centric products and services. Its end markets include hotels, commercial buildings, healthcare facilities, etc. Yunji is the largest company in the hospitality robotics-based AI agent for hotel scenario in China. Currently, it has delivered over 60k units of robotics to over 40k hotels in China.

Management targets for >50% p.a. YoY revenue growth in 2026/27E. Looking ahead, the company believes future revenue growth drivers include 1) penetration into more hotels; 2) penetration into other markets (i.e., service apartments, high-end communities etc.); 3) more diversified types of services; and 4) replacement and upgrade demand (replacement cycle of five years). Meanwhile, management targets to lift the percentage of revenue from AI-agent which has higher margin in the longer term (vs. hardware).

Exhibit 5: Yunji's service robots

Yunji is the largest company in the hospitality robotics-based AI agent for hotel scenario in China

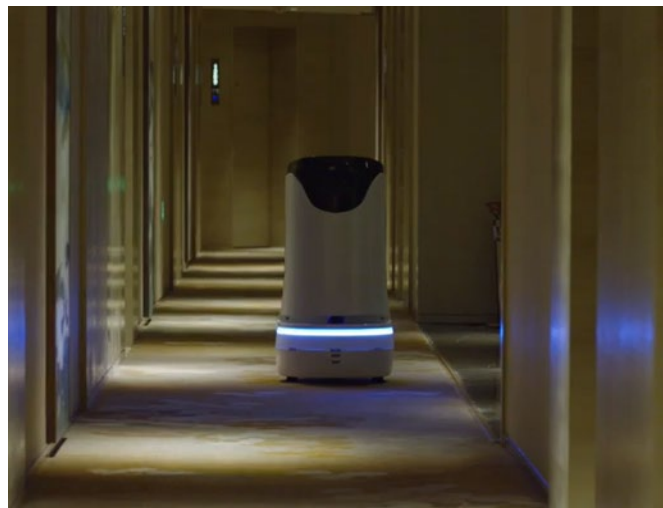


Source: BofA Global Research, Yunji

BofA GLOBAL RESEARCH

Exhibit 6: Yunji's intelligent delivery robot - RUN

RUN can autonomously deliver items to hotel customers



Source: Yunji

BofA GLOBAL RESEARCH

Key takeaways – key component suppliers

Linkerbot (private)

Linkerbot is a leading Chinese robotics company focused on developing high-DoF dexterous hands and embodied AI systems, leveraging advanced actuation, force control, and perception technologies to enable human-like manipulation, precise grasping, and complex task execution in real-world environments. It has 80% global market share in high-DoF (>20 DoF) dexterous hands.

Linkerbot dexterous hands cover all three major technology routes including tendon drive, direct drive and bar linkage. Currently, the ASP of its dexterous hand (O6 Lite) can reach as low as RMB3999 (USD590), much lower than its peers, due to (1) scale effect (>10k unit shipment in 2025); (2) improved production process (it can produce dexterous hand with its own robotic arms); and (3) modularized design. It has internal design of major components including coreless motor, mini screw and reducers. It has also established multiple data collection centers to collect manipulation-related data. Management sees dexterous hand companies are well-positioned in manipulation-related data collection, as the majority of humanoid robot manipulation activities are related to dexterous hand.

Exhibit 7: Linkerbot's Linker Hand L20

Linkerbot has 80% global market share in high-DoF dexterous hands

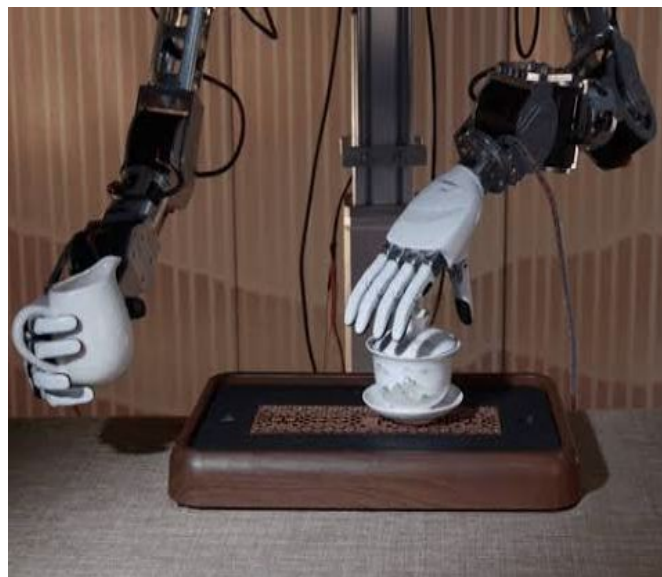


Source: Linkerbot

BofA GLOBAL RESEARCH

Exhibit 8: Linkerbot's dexterous hand performing pick-and-place

Linkerbot's dexterous hand has achieved fast cost reduction over years



Source: Linkerbot

BofA GLOBAL RESEARCH

Orbbec (688322 CH)

Orbbec is one of the leading companies of robot vision system in China. It has developed full-stack technology capability in robot vision system, including sensory chips, visual sensors (structural light camera, indirect time-of-flight (iToF), stereo camera, light detection and ranging (LiDAR)), vision analysis algorithms and software packages. The company has a solid market position in robotic vision systems in China, especially for structural light camera and iToF. Notably, it has >70% market share in the 3D vision system of service robots in China.

Management expects Orbbec's robotics-related business to become a core growth driver, with humanoid and service robots-related business currently contributing c.30% of its total revenue and potentially exceeding 50% of total revenue within two years. A typical humanoid robot uses 1-2 red, green, blue + dept (RGBD) cameras, with some configurations adopting up to 4, representing content value of roughly RMB2,000-3,000 per robot (assuming 2 cameras per robot). Management expects the cost of robotic cameras to decline 50% in the coming decade, thanks to improved product design and scale effect. Meanwhile, the company is also expanding its data-collection ecosystem through RGBD camera, stereo-vision and wearable solutions for teleoperation and Egocentric data collection.

Exhibit 9: Orbbec's binocular structured light camera

Orbbec is one of the leading players of robot vision system in China



Source: Orbbec

BofA GLOBAL RESEARCH

Exhibit 10: Orbbec's Egocentric data collection module

Orbbec has launched wearable solutions for teleoperation and Egocentric data collection



Source: Orbbec

BofA GLOBAL RESEARCH

PaXini (private)

PaXini is a leading Chinese company specializing in multidimensional tactile sensing technology, with products including tactile sensor, torque sensor, 6D force and torque sensor, dexterous hand (DexH13/DexH5), wheeled humanoid robot (TORA) and dataset/data collection solutions. PaXini adopts a proprietary 6D Hall-effect tactile sensing technology that provides high-precision multidimensional force and tactile perception, with precision of up to 0.01N and 0.1mm, enabling robots to accurately detect contact, pressure, torque, and slip during manipulation. The company has achieved significant cost reduction of tactile sensors from RMB100k/unit to RMB200/unit in 5-6 years, due to continuous improvement in product design and scale effect.

PaXini has announced to establish five data collection factories in Tianjin, Suqian, Wuhan, Zigong and Ganzhou, which can generate >10bn pieces of data per year upon completion, with daily data output equivalent to the volume of data generated during a human's whole life. The five data collection factories cover different scenarios, including logistics, industrial manufacturing and retail. PaXini's data collection is mainly based on UMI and Egocentric methods, which can achieve 3-6x higher efficiency relative to teleoperation, and can provide high-quality data to VTLA models. The company can achieve external sales of its multimodal embodied AI data via its online data platform, OmniSharingDB.

Exhibit 11: PaXini's TORA robot demonstrating item sorting

PaXini specializes in multidimensional tactile sensing

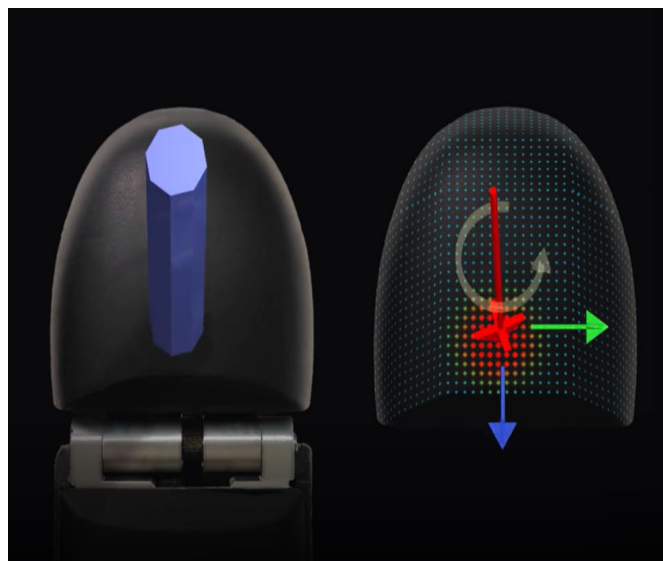


Source: BofA Global Research, PaXini

BofA GLOBAL RESEARCH

Exhibit 12: PaXini's tactile sensor

PaXini's tactile sensor can collect multidimensional data with multimodal perception



Source: PaXini

BofA GLOBAL RESEARCH

Appendix

Exhibit 13: The global humanoid robot supply chain (2026)

Greater China companies play a pivotal role in the humanoid robot supply chain

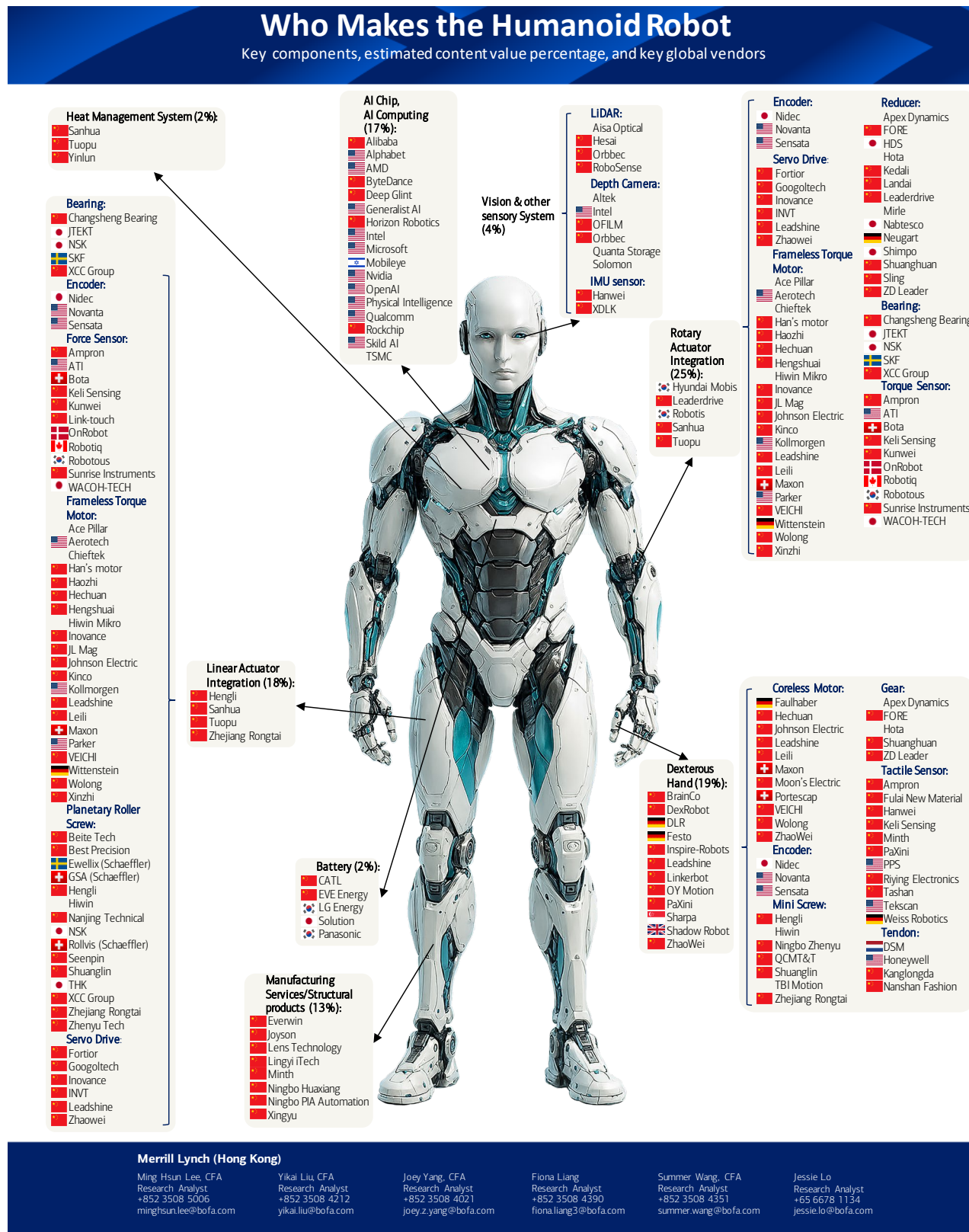


Exhibit 14: Greater China humanoid robot supply chain companies

Greater China humanoid robot component manufacturers play a pivotal role

Company	Ticker	Market cap (\$mn)	6-month ADTV (\$mn)	Major product
Humanoid/service robot				
XPeng	9868 HK	11,175	129	Humanoid robot
Shangwei New Material	688585 CH	9,984	109	Composite materials, humanoid robot
UBTECH	9880 HK	5,801	108	Humanoid robot
Estun	002747 CH	4,900	288	Industrial robot, humanoid robot
OneRobotics	6600 HK	1,788	7	Service robot, humanoid robot
Dobot	2432 HK	1,296	14	Collaborative robot, humanoid robot
Yunji	2670 HK	1,295	1	Service robot
Galaxea AI	NA	NA	NA	Humanoid robot, wheeled robot
Actuator integration				
Sanhua	002050 CH/2050 HK	23,565	56	Rotary actuator, linear actuator, heat management
Hengli Hydraulic	601100 CH	21,308	162	Linear actuator, ball screw, planetary roller screw, dexterous hand actuator
Tuopu	601689 CH	13,118	328	Rotary actuator, linear actuator, heat management
Servo motor				
Inovance	300124 CH	25,671	376	Servo motor, servo controller
Wolong Electric	600580 CH	8,485	279	Coreless motor, robot actuator
Haozhi Industrial	300503 CH	3,442	251	Brushless DC motor, harmonic reducer, encoder
Leadshine	002979 CH	2,848	109	Brushless DC motor, coreless motor, micro servo system
Jiangsu Leili	300660 CH	2,374	48	Brushless DC motor, coreless motor, linear motor
Johnson Electric	179 HK	2,358	24	Micro servo motor
VEICHI Electric	688698 CH	1,592	57	Brushless DC motor, coreless motor
Kinco Automation	688160 CH	1,547	49	Brushless DC motor
Jiecang Linear Motion	603583 CH	1,445	22	Coreless motor, linear motor
Hiwin Mikrosystem	4576 TT	811	21	Servo motor, servo controller
Hechuan	688320 CH	730	25	Brushless DC motor, coreless motor
Chieftek	1597 TT	409	12	Linear motor
Ace Pillar	8374 TT	286	4	Servo motor, servo controller
Dexterous hand				
Hengli Hydraulic	601100 CH	21,308	162	Linear actuator, ball screw, planetary roller screw, dexterous hand actuator
Moon's Electric	603728 CH	3,284	75	Dexterous hand, brushless DC motor, coreless motor
ZhaoWei	003021 CH	3,053	74	Dexterous hand, micro-drive system
Nanshan Fashion	300918 CH	803	18	Tendon
Kanglongda	603665 CH	532	14	Tendon
Linkerbot	NA	NA	NA	Dexterous hand
Reducer				
Leaderdrive	688017 CH	9,918	525	Harmonic reducer
Kedali	002850 CH	8,226	167	Harmonic reducer
Shuanghuan	002472 CH	4,926	177	Planetary reducer, harmonic reducer
Sling	301550 CH	4,345	134	Harmonic reducer, bearing
ZD Leader	002896 CH	2,418	128	Planetary reducer, harmonic reducer, robot actuator
Mirle Automation	2464 TT	1,258	60	Harmonic reducer
Apex Dynamics	4583 TT	1,242	4	Planetary reducer
FORE	301368 CH	778	21	Planetary reducer, harmonic reducer
Hota Industrial	1536 TT	442	4	Planetary reducer
Bearing				
Zhejiang XCC Group	603667 CH	3,215	282	Bearing, bearing ring, rolling element, planetary roller screw components
Zhejiang Changsheng Bearing	300718 CH	3,040	92	Bearing, ball screw
Ball screw/planetary roller screw				
Hengli Hydraulic	601100 CH	21,308	162	Linear actuator, ball screw, planetary roller screw, dexterous hand actuator
Hiwin	2049 TT	4,032	62	Ball screw, harmonic reducer
Zhejiang Rongtai	603119 CH	3,957	173	Planetary roller screw
Zhenyu Technology	300953 CH	3,815	174	Planetary roller screw, motor core, robot actuator structural parts
Beite Technology	603009 CH	2,259	68	Planetary roller screw
Shuanglin	300100 CH	2,006	53	Ball screws
Qinchuan Machine Tool	000837 CH	1,650	43	Reducer/ball screw grinding machine
Best Precision	300580 CH	1,635	49	Planetary roller screw
TBI Motion	4540 TT	208	12	Ball screw
Vision system				
Orbbec	688322 CH	6,999	228	3D vision, Time-of-flight (ToF) camera, LiDAR
OFILM	002456 CH	3,762	141	Optical machine modules
Hesai	HSAL US	3,033	38	LiDAR
RoboSense	2498 HK	1,386	22	3D vision, LiDAR
Aisa Optical	3019 TT	1,193	30	Lens, camera module

Exhibit 14: Greater China humanoid robot supply chain companies

Greater China humanoid robot component manufacturers play a pivotal role

Company	Ticker	Market cap (\$mn)	6-month ADTV (\$mn)	Major product
Solomon Technology	2359 TT	746	18	3D vision
Quanta Storage	6188 TT	619	5	3D vision, collaborative robot
Altek	3059 TT	385	8	3D vision, vision AI chip
Sensor				
XDLK	688582 CH	2,503	57	IMU sensor
Keli Sensing Tech	603662 CH	2,239	76	Torque sensor, 6-axis force sensor
Hanwei Electronics	300007 CH	1,705	88	Temperature sensor, pressure sensor, inertial measurement unit (IMU)
Shenzhen Ampron	301413 CH	1,322	66	Torque sensor, 6-axis force sensor
PaXini	NA	NA	NA	Tactile sensor, torque sensor, dexterous hand, wheeled robot, dataset
AI Chip, AI Computing				
TSMC	2330 TT	1,923,564	2,928	Semiconductor manufacturing
Rockchip	603893 CH	13,381	370	Processor chips
Horizon Robotics	9660 HK	10,307	127	Processor chips
SenseTime	20 HK	7,694	118	AI algorithm and solutions
Cloudwalk	688327 CH	1,685	61	AI algorithm application, human-machine collaborative operating system
Deep Glint	688207 CH	752	36	AI algorithm and solutions
Servo Drive				
Fortior	688279 CH/1304 HK	2,637	56	Servo drive controller chip
Googoltech	301510 CH	1,603	52	Servo drives
INVT	002334 CH	829	45	Servo drives
Battery				
CATL	3750 HK	273,220	308	Battery module
EVE Energy	300014 CH	18,054	644	Battery module
Azure	002245 CH	4,490	287	Battery module
Heat Management System				
Sanhua	002050 CH/2050 HK	23,565	56	Rotary actuator, linear actuator, heat management
Yinlun	002126 CH	5,235	180	Heat management products
Manufacturing Services/Structural products				
Lens Technology	300433 CH/6613 HK	26,333	57	Manufacturing services, 3C components
Lingyi iTech	002600 CH	15,905	397	Manufacturing services
Everwin	300115 CH	5,686	246	Structural products, 3C components
Joyson	699 HK	4,663	6	Head module
Minth	425 HK	4,210	25	Face mask
Ningbo Huaxiang	002048 CH	2,346	80	Manufacturing services, PEEK material
Ningbo PIA Automation	688306 CH	1,671	19	Manufacturing services

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH



Disclosures

Important Disclosures

BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.

Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.

The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.

Officers of BofA or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.

Refer to [BofA Global Research policies relating to conflicts of interest](#).

"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.

Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:

BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofamli.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSF); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority. This information has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSF; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the [Electronic Communications Disclaimers](#) for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives,



financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are largely unregulated. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities that may have been recommended in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQDatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.